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Multi-Task Self-Supervised Learning for Automated Measurement of Left Ventricular Ejection Fraction in Echocardiography

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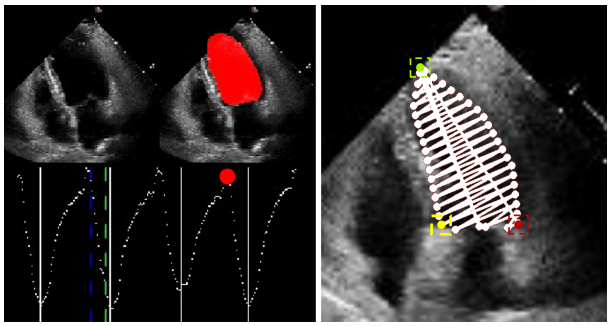
***Abstract*—Accurate segmentation and landmark detection are essential for medical image analysis. However, existing methods often struggle with generalization and robustness, particularly under sparse annotation conditions. This study introduces SSP-MAEFNet, a novel self-supervised framework designed to address these limitations. The model integrates a lightweight backbone network, LR-ASPP-MobileNetV3, for efficient feature extraction, along with a Deep Separable Task-Specific (DSTS) module that mitigates feature entanglement between segmentation and landmark detection tasks. By leveraging a self-supervised random masking and reconstruction strategy, the framework effectively captures temporal and spatial patterns, enhancing its generalization across datasets with minimal supervision. Extensive experiments on the EchoNet-Dynamic dataset demonstrate that SSP-MAEFNet achieves state-of-the-art performance, with a Dice score of 93.54% and an OKS of 81.30%, outperforming existing models in both tasks. Its robust performance under sparse annotation highlights its potential for clinical integration, especially in resource-constrained environments.**

***Index Terms*—Self-Supervised Learning, Deep Learning, Medical Image Segmentation, Landmark Detection**

I. Introduction

In clinical practice, the left ventricular ejection fraction (LVEF) is most commonly assessed using the Simpson biplane method. [1]. This process requires manual delineation of the left ventricular endocardium in key frames from apical two-chamber echocardiographic views (A2C) and apical four-chamber echocardiographic views (A4C), as illustrated in Fig. 1. However, this work is time-consuming and prone to variability between practitioners. Therefore, it is necessary to develop an automated method for calculating LVEF that is consistent with the methods used in the clinical settings.

Currently, the main techniques for calculating the left ventricular ejection fraction (LVEF) are based on image segmentation. However, a major limitation of these segmentation-based methods is their inability to directly identify the apical landmark and the position of the mitral annulus from the segmentation results. This limitation hinders the direct application of the Simpson method to calculate the left ventricular volume, thereby affecting the accuracy and reliability of cardiac function assessment.



(a) (b)

Fig. 1. The process of clinicians processing echocardiograms. (a) Manual selection of key frames; (b) Annotation of the left ventricular endocardial border. The white dots represent points manually annotated by clinicians. The yellow box indicates the septal mitral annulus, the red box indicates the lateral mitral annulus, and the green box indicates the cardiac apex.

To address this challenge and improve the quality of cardiac function assessment, we propose a self-supervised framework with task-specific feature learning for image segmentation and landmark detection. This framework is structured into two distinct stages. In the initial stage, a self-supervised contrastive model is developed. By employing random masking and reconstruction strategies, we train an encoder based on the MobileNetV3 architecture [2], designed to capture periodic motion patterns and temporal information from both adjacent and non-adjacent frames. The trained encoder's weights are subsequently transferred to the second stage, where they are utilized within a task-specific decoder tailored for segmentation and landmark detection tasks. Given that only two frames per video are annotated, a weakly supervised approach is employed to generate pseudo-labels, which are then used to analyze full video sequences. To effectively address feature entanglement in multi-task learning, we develop a module that is integrated into the decoder, which is Deeply Separable Task-Specific (DSTS). Finally, Ejection Fraction could be calculated

using segmentation and landmark detection results. The main contributions of this study are as follows.

- We propose SSP-MAEFNet, a multi-task self-supervised framework that jointly performs segmentation and landmark detection to enable automated LVEF measurement from echocardiographic videos.
- A novel DSTS module is designed to disentangle task-specific features and mitigate interference in multi-task learning, enhancing segmentation and landmark detection performance.
- The framework leverages self-supervised pretraining with random masking, enabling robust spatial-temporal representation learning under sparse annotations, while its lightweight design supports deployment in point-of-care and resource-constrained settings.

II. Related Work

A. Self-Supervised Learning (SSL)

Self-supervised learning (SSL) leverages proxy tasks to learn feature representations from unlabeled data, reducing reliance on manual annotations. In echocardiography, contrastive pretraining [3], temporal modeling [4], and SAM-based integration [5] have improved segmentation under sparse labels. Beyond single-task learning, recent studies have explored multi-task SSL frameworks for shared representation learning [6], [7]. Building on these advances, our work integrates self-supervised pretraining with multi-task modeling to jointly perform segmentation and landmark detection in echocardiographic videos.

B. Segmentation of Left Ventricle

Left ventricular segmentation is essential for LVEF calculation in cardiac function analysis. To enhance segmentation accuracy, researchers have developed U-Net variants [8], attention-based models [9], and Transformer architectures such as SegFormer [10] and DeepLabv3+ [11]. EchoEFNet [1] and EchoGraph [12] introduced multi-task and spatiotemporal modeling for improved performance in echocardiographic videos. In low-annotation settings, SepXception [13] and MemSAM [14] advanced weakly supervised segmentation using novel loss functions and unlabeled data mining. These efforts reflect a growing trend toward accurate and annotation-efficient segmentation for cardiac imaging.

C. Landmark Detection

Anatomical landmark detection is critical for cardiac function assessment, particularly for locating the mitral annulus and apex when applying the Simpson biplane method to compute LVEF. In medical imaging, convolutional neural networks (CNNs) have shown strong performance in keypoint localization tasks. Shen et al. [15] preserve spatial detail and are widely adopted for accurate landmark detection. Additionally, Li et al. [16] reformulated keypoint estimation as a classification task, reducing quantization errors and improving detection precision. These methods highlight the importance of high-resolution and task-aware representations in anatomical landmark detection.

III. Framework

A. Overview of SSP-MAEFNet

SSP-MAEFNet integrates self-supervised pretraining, weakly supervised learning, and multitask feature extraction into a unified framework, designed to enhance segmentation accuracy and landmark detection while ensuring clinical relevance. The workflow begins with an input echocardiogram video, where a random mask reconstruction task enables the model to learn robust spatio-temporal features during self-supervised pretraining. The pretrained weights are then utilized in a weakly supervised dual-branch network, which incorporates the Deeply Separable Task-Specific (DSTS) module to resolve feature entanglement and produce both segmentation maps and landmark detection outputs. Finally, the ejection fraction is calculated using the Simpson method, integrating multitask output to ensure clinically accurate and interpretable results. This end-to-end design aligns with real-world clinical workflows, offering improved automation and reliability in echocardiogram analysis.

B. Self-Supervised Pre-training Learning

In echocardiographic data, annotations are typically available only for two key frames. For each frame, a random mask is applied with a mask ratio of 50%, setting the pixel values of the masked regions to 0. This results in a processed dataset V_m, which is then input into the algorithm to extract features from the remaining visible pixels. Based on these extracted features, the network reconstructs the original video V.

The network is optimized by minimizing the difference between the original and reconstructed videos, which is calculated using the loss function of mean squared error (MSE) as follows:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N}\sum_{n=1}^{N} \text{MSE}(V^n, \text{Net}(V_m^n)), \tag{1}$$

where V^n represents the original frame, V_m^n represents the masked input frame, and N is the total number of video samples. This stage allows the model to learn robust spatio-temporal features from unlabeled data.

C. Deeply Separable Task-Specific (DSTS) Module

As shown in Fig. 3, the DSTS module uses the correlation matrix between semantic segmentation and landmark detection tasks to extract modal information from shared features. The input data are processed with deep separable convolution (stride 2) and group convolution to generate Query (Q) and Key (K) for self-attention. Separable convolution captures local features and reduces computational complexity, while group convolution reduces parameters and mitigates overfitting.

The features are flattened into a matrix of size $R^{\mathrm{N}\times\mathrm{C}}$, where $\mathbf{N} = \mathbf{H} \times \mathbf{W}$. After passing through LayerNorm (LN) and linear projection (LP), the Query (Q), Key (K), and Value (V) for each task branch are generated. The operation is expressed as follows.

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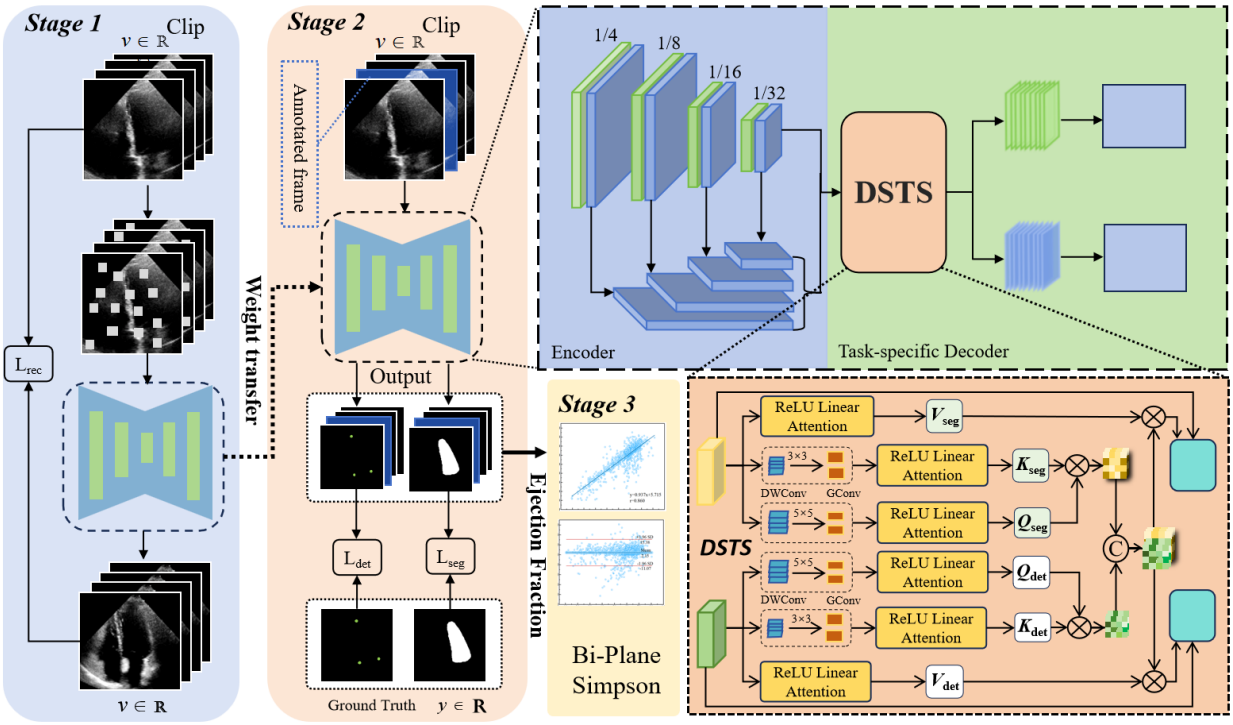


Fig. 2. Overview of SSP-MAEFNet: Stage.1. Self-supervised pre-training with random masking and pixel block reconstruction in echocardiographic videos. Stage.2. Multi-task learning with weak supervision, using labeled frames for loss computation and pseudo labels for unlabeled frames to analyze full video sequences. Stage.3. Ejection fraction calculation using the biplane Simpson method.

$$
\begin{aligned}
Q &= W_q \times \mathrm{LP}\left(\mathrm{Flat}\left(\nu_{s\times 5}(F)\right)\right), \\
\mathbf{K} &= W_k \times \mathrm{LP}\left(\mathrm{Flat}\left(\nu_{3\times 3}(F)\right)\right), \\
V &= W_\nu \times \mathrm{LP}\left(\mathrm{Flat}(F)\right),
\end{aligned} \tag{2}
$$

Subsequently, the spatial global correlation weight matrix $\mathbf{W}$ is obtained through a dot product operation, and its calculation formula is given by:

$$
W = \frac{Q \times K^T}{\sqrt{d_k}}, \quad W \in \mathbb{R}^{(\lfloor \frac{H-5}{2} \rfloor+1)\times(\lfloor \frac{W-5}{2} \rfloor+1)\times C} \tag{3}
$$

The weight matrix $\mathbf{W}$ from both tasks is concatenated, followed by a 1×1 convolution to align dimensions and share task-specific parameters. Then, $\mathbf{W}$ is multiplied by the value matrix $\mathbf{V}$ to obtain task-specific features. The formula for this calculation is given by:

$$
\begin{aligned}
W &= \nu_{1\times 1}(\mathrm{Concat}(W_{\mathrm{seg}}, W_{\mathrm{det}})) \\
x &= \mathrm{LP}(\mathrm{ReLU}(W) \times V) \\
F^{'} &= \mathrm{MLP}(\mathrm{LN}(x+F)) + (x+F)
\end{aligned} \tag{4}
$$

F is the input module feature, and F' is the module output feature.

D. Estimation of the LVEF based on Simpson's Method

In this study, the Simpson biplane method was used to calculate the LVEF. As shown in Fig. 4.

The curve of the left ventricle area is generated from ultrasound segmentation, which identifies the end diastolic (fED) and end systolic (fES) frames. The internal diameter

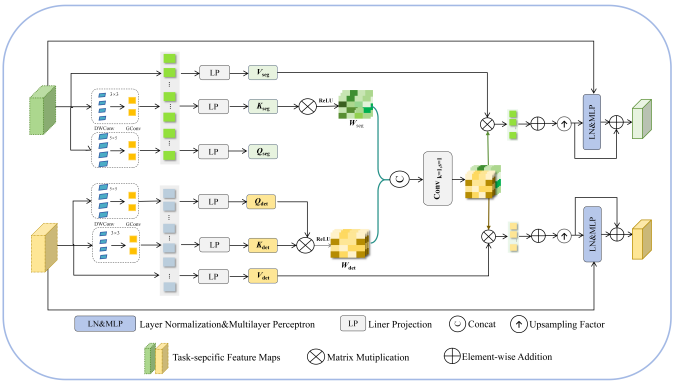


Fig. 3. DSTS module: input data undergoes deep separable and group convolutions to generate Query and Key for the self-attention mechanism. A correlation matrix from segmentation and landmark detection tasks extracts shared features, producing task-specific outputs for both tasks.

of the left ventricle (LVID) is determined by connecting the markers on the mitral valve ring to the apex. The left ventricle is divided into 20 discs along the LVID, and LVEDV and LVESV are calculated using formula 5. LVEF is then derived from formula 5, with a_i, b_i, and L representing the diameters of the disc and the length of the LVID. To simplify the calculation, we set a_i=b_i.

$$
V = \frac{\pi}{4} \cdot \frac{L_{\max}}{20} \sum_{i=1}^{20} a_i b_i, \; LVEF = \frac{V_{\mathrm{ED}} - V_{\mathrm{ES}}}{V_{\mathrm{ED}}} \times 100\% \tag{5}
$$

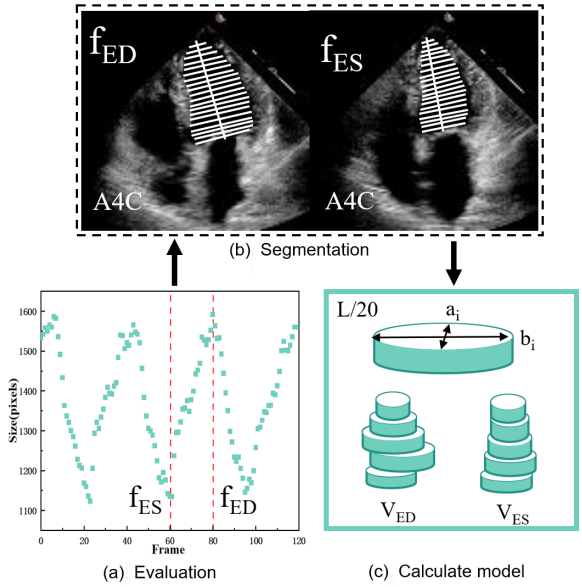


Fig. 4. Workflow for LVEF Calculation (a) Derive the left ventricular area curve from the A4C view to identify the ED and ES frames. (b) Divide the segmentation results into disks based on anatomical landmarks. (c) Calculate LVEDV and LVESV using the disk-based model.

E. Loss Function Design

The SSP-MAEFNet training process is divided into two stages: the pre-training stage and the fine-tuning stage. Each stage is optimized with a task-specific loss function to ensure effective learning.

Stage 1: Self-Supervised Pre-Training. In the pre-training stage, the model is optimized using only the reconstruction loss $\mathcal{L}_{\text{rec}}$.

Stage 2: Fine-Tuning. In the fine-tuning stage, the framework integrates multi-task learning to optimize segmentation, landmark detection, and reconstruction tasks simultaneously. The total loss function is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{det}}\mathcal{L}_{\text{det}} + \lambda_{\text{seg}}\mathcal{L}_{\text{seg}}, \tag{6}$$

where:

- $\mathcal{L}_{\text{rec}}$: Reconstruction loss (MSE) for maintaining consistency in spatio-temporal features.
- $\mathcal{L}_{\text{det}}$: Landmark detection loss, defined as:

$$\mathcal{L}_{\text{det}} = \frac{1}{2N}\sum_{i=1}^{N}\left[(x_{\text{pred}}^{(i)} - x_{\text{true}}^{(i)})^2 + (y_{\text{pred}}^{(i)} - y_{\text{true}}^{(i)})^2\right]. \tag{7}$$

- $\mathcal{L}_{\text{seg}}$: Binary cross-entropy loss for segmentation tasks, defined as:

$$\mathcal{L}_{\text{seg}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \ln(\hat{y}_i) + (1-y_i)\ln(1-\hat{y}_i)\right]. \tag{8}$$

The hyperparameters λ_{rec}, λ_{det}, and λ_{seg} are used to balance the contributions of each task.

IV. EXPERIMENT

A. Datasets and Configurations

EchoNet-Dynamic: This dataset, among the largest annotated medical video datasets, includes 10,030 112x112 videos with expert-marked endocardial edges at end-systole and end-diastole [17]. Ventricular volumes are estimated using traced areas represented by 32 paired coordinates, along with core cardiac function indicators.

Pre-training: A new dataset is generated by randomly masking EchoNet-Dynamic, and the model is trained for 100 epochs with a batch size of 16. The hyperparameters include an initial learning rate of 0.001 (with exponential decay), weight decay of 0.0005 and Adam optimizer.

Fine-tuning: We fine-tune the model for 20 rounds with a batch size of 8 using the Adam optimizer. The experiments are carried out in Python 3.8 with PyTorch 1.13 on 8 NVIDIA RTX-4090 GPUs.

B. Segmentation Results

The segmentation results in Table I show that our model balances high accuracy and computational efficiency, making it suitable for practical applications. Fig. 5 shows that it is challenging to distinguish the endocardial and fibrous regions of the left ventricle.

TABLE I
SEGMENTATION RESULTS IN ECHONET-DYNAMIC DATASET

Model	DSC(↑)	HD(↓)	Params(M)(↓)	Flops(G)(↓)
U-Net [18]	92.48	4.501	14.705	10.434
nnU-Net [8]	92.83	4.379	19.162	14.277
EchoGraph [12]	91.62	4.685	27.580	21.538
MemSAM [14]	93.15	4.028	138.863	109.594
SegFormer [10]	90.76	5.583	3.643	11.471
SepXception [13]	92.88	4.475	55.832	85.684
TransBridge [19]	91.02	5.213	**3.490**	8.512
DeepLabv3 [11]	90.17	5.429	19.065	0.863
simLVseg [4]	93.32	3.795	125.768	7.31
EchoNet [17]	92.23	4.253	39.647	10.619
EchoEFNet [1]	92.29	4.643	32.429	1.472
SSP-MAEFNet	**93.54**	**3.512**	5.007	**0.246**

C. Landmark Detection Results

Table II shows that, despite high-accuracy models that typically require more resources, our optimized architecture with a lightweight MobileNetV3 backbone reduces computing costs without sacrificing performance. Fig. 6 shows the difficulty in detecting landmarks on echocardiograms, particularly in blurred areas such as the lateral mitral annulus and the cardiac apex. Our model outperforms others in landmark detection accuracy.

D. Comparative Experiment of Feature Extraction Network

The experimental results in Table III show that MobileNetV3 achieves the best segmentation performance in EchoNet-Dynamic dataset. As an encoder, MobileNetV3 has the lowest HD of 3.139, indicating that the segmentation edges of the left ventricle are clear and continuous.

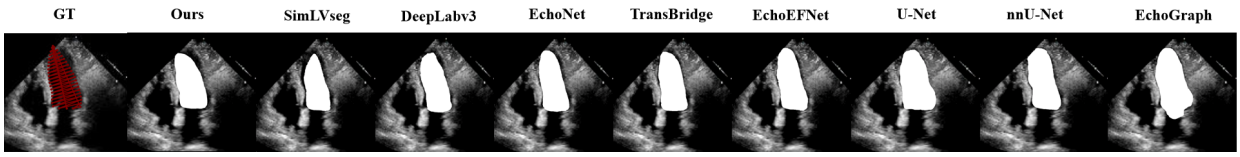


Fig. 5. The segmentation results of different models.

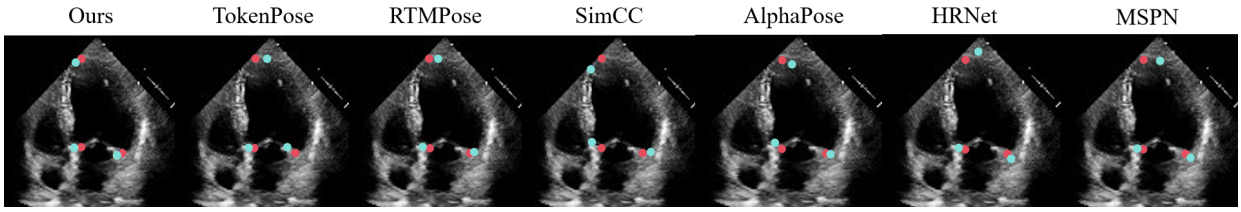


Fig. 6. The landmark detection results of different models. The red points are annotations, and the blue points are predicted values

TABLE II
LANDMARK DETECTION RESULTS IN ECHONET-DYNAMIC DATASET

Model	OKS (%)(↑)	Params (M)(↓)	FLOPs (G)(↓)
HRNet [15]	74.88	62.725	3.732
MSPN [16]	76.39	27.541	1.56
TokenPose [20]	76.70	5.193	1.434
AlphaPose [21]	80.30	28.326	3.507
SimCC [22]	76.00	25.082	2.285
RTMPose [23]	80.90	28.639	1.065
SSP-MAEFNet	**81.30**	**5.007**	**0.246**

TABLE III
MULTI-TASK RESULTS WITH DIFFERENT BACKBONES

Backbone	Dice (%)(↑)	HD(↓)	OKS(↑)
MobileNet [24]	73.58	5.646	0.726
ResNet50 [25]	90.72	4.163	0.715
SwinTransformer [26]	89.03	4.472	0.691
PoolFormer [27]	90.13	5.034	0.740
DeiT [28]	84.96	5.931	0.684
EfficientFormer [29]	91.25	3.662	0.760
ReoViT [30]	91.44	3.735	**0.782**
MobileNetV3 [2]	**91.56**	**3.139**	0.778

TABLE IV
IMPACT OF DSTS MODULES ON DIFFERENT BACKBONE NETWORKS

Backbone	DSTS	Dice (%)(↑)	HD(↓)	OKS(↑)
Mobi1eNet	✓	80.33	4.902	0.754
ResNet50	✓	92.05	3.791	0.742
SwinTransformer	✓	90.94	4.083	0.728
PoolFormer	✓	89.44	5.216	0.731
DeiT	✓	86.62	5.428	0.717
EfficientFormer	✓	91.10	**3.645**	0.764
ReoViT	✓	91.89	3.812	0.790
MobileNetV3	✓	**92.47**	4.024	**0.795**

TABLE V
DSTS AND PRE-TRAINING EFFECTS

Model	Dice	HD	OKS	params (M)	GFlops
Base	91.56	4.139	0.778	4.953	0.227
Base+DSTS	92.47	4.024	0.795	5.007	0.246
Base+pretrain+DSTS	93.54	3.512	0.803	5.007	0.246

TABLE VI
ANALYSIS OF PREDICTED AND TRUE VALUES OF EDV, ESV, AND LVEF

Dataset	Indice	Corr	LOA	PELOA	ICC
EchoNet-Dynamic	EDV	0.943	21.057	23.055	0.943
	ESV	0.976	14.853	34.066	0.954
	LVEF	0.86	13.453	23.192	0.787

E. Ablation Experiment

The DSTS that we propose shows certain performance improvements in various feature extraction networks, as shown in Table IV. Table V shows that DSTS and pretraining improve model performance. Despite slight increases in parameters (5.007M) and complexity (0.246 GFlops), the improvements are valuable for practical applications. Its wide applicability and strong generalization make it valuable for complex tasks such as medical image analysis.

F. Clinical Reliability Analysis

As can be seen in Table VI and Fig. 7, the ESV has the closest linear relationship. The LVEF correlation coefficient is relatively weak. The possible reason is that the variation range of the end-systolic frame in the data set is significantly smaller than that of the end-diastolic frame. The LOA of the ESV is relatively small, indicating that in the prediction of end systolic volume, the difference between the predicted value and the true value is relatively concentrated.

Since LVEF is calculated based on EDV and ESV, any error between these two values increases the error in LVEF. Specifically, the LVEF ICC (0.787) is low, indicating that its prediction reliability and consistency are not as strong as those of EDV and ESV. However, the estimated value within the confidence interval 95% of the ICC ranges from 0.75 to 0.9, suggesting that the clinical reliability is good and meets the needs of practical applications.

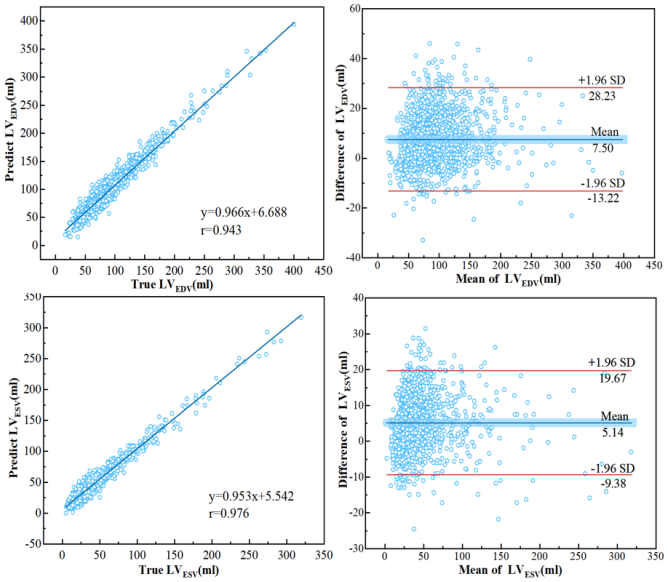


Fig. 7. Distribution correlation and Bland-Altman plots diagram of predicted and true values of EDV, ESV, and LVEF

V. Conclusion

We present SSP-MAEFNet, a multi-task self-supervised framework for automated LVEF measurement from echocardiographic videos. By jointly performing segmentation and landmark detection, the model enables accurate cardiac function assessment under sparse annotations. Its two-stage design—self-supervised pretraining and task-specific decoding—achieves strong generalization and state-of-the-art performance on the EchoNet-Dynamic dataset. With a lightweight structure and minimal labeling requirements, SSP-MAEFNet is well suited for point-of-care deployment. Future work will explore additional cardiac views and enhance robustness through domain adaptation and clinical validation.

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